

Adroit Technologies

Protocol Driver

Name	Mitsubishi Ethernet 2 nd Generation (Q, L, iQR and iQF)
DLL	D_MIT_ETH



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Contents

1. Overview	4
1.1. Upgrading from classical MITQJE71 Driver	4
2. Wiring	5
3. Device Configuration	7
3.1. Device Configuration	7
3.1.1. Error Handling	7
3.1.2. Latency	7
3.1.3. Ethernet	7
3.1.4. PLC Details	8
3.1.5. Test	9
3.1.6. Driver Send Watchdog	9
3.1.7. Validation	10
3.1.8. Data Ranges for Selected PLC Model (*WARNING: Denoted in DECIMAL below)	10
4. Supported Addresses	11
4.1. Mitsubishi Q series (System Q)	11
4.2. Mitsubishi QnA series PLC	12
4.3. Mitsubishi L series PLC data	12
4.4. Mitsubishi R series PLC data	13
4.5. Mitsubishi FX5 series PLC data	13
4.6. Omitted	14
4.7. Non-bit Device Writes	14
5. Protocol Definition and Considerations	15
5.1. PLC software and Heartbeat code	15
5.1.1. PLC Side Communication setup	15
5.2. Using GX Works2 (Used for All except R Series CPU)	15
5.2.1. Settings for CPU without an on-board Ethernet Connection (Q02)	15
5.2.2. Settings for CPU with an on-board Ethernet Connection (Q03UDE)	17
5.3. PLC Setup with PLC-SCADA Watchdog	19
5.3.1. Initial Settings	19
5.3.2. PLC Watchdog Program	19
5.4. Use the GX Works3 Software (For R and FX5 Series CPU's)	20
5.5. The MC Protocol 3E Frame [Binary]	22
5.5.1. Request-Frame Format	22
5.5.2. Response-Frame Format	22
5.5.3. Text Portion of the 3E/3C Frame	22
5.5.4. Device Codes	23
6. General Notes and Troubleshooting	24
6.1. Contact Details	24
6.2. Adroit Technologies Knowledge Base	24
6.3. Troubleshooting Steps	24
6.4. Error Message Reference:	25
7. Revisions	26
7.1. Document Revision	26
7.2. DLL Revision	26

1. Overview

This driver supports the Mitsubishi 3E Frames protocol, with communication to the Q/QnA/L/R and FX5 Series CPU's.

The 2nd Generation driver allows for the following features:

- a) Improved performance due to optimizations on mixed I/O to the same protocol area being transacted only once removing redundant transactions.
- b) Support for multiple channels, where due scan tasks are balanced over available channels, reads and writes are interlaced to prevent starvation.
- c) Demand scanning support, where read only I/O that is not on a mimic / graphic form and not logged or alarmed and not output enabled will be removed from the scanning system, the applicable tasks will dynamically be reduced in size and can merge into fewer scan tasks.

For improved network performance and reliability, it is recommended that you install 2 network adapters into your PC. Use the first adapter for Adroit to Adroit and all other Windows connectivity. The first adapter may have all the default installation bindings. Use the second adapter for Adroit to PLC connectivity and bind only to the TCP/IP protocol for PLC communications.

1.1. Upgrading from classical MITQJE71 Driver

The scanning addresses are identical.

If using the MITQJE71 driver, remove all of the old Device Agents, and create them again with the same names using the Q driver. Take care to note settings like the IP, port address and watchdog addresses. On the A-Series driver the Decimal or Hex notation used is important and does change over in this Q/QnA driver. Names of I/O types stay the same so far possible, this means that scanned I/O types existing in the A-Series driver that have different names (prefixes) in the Q Driver are promoted automatically to their new names. After saving, the WGP file cannot be loaded on the old driver again without the possibility of losing some scan-points.

2. Wiring

As per Ethernet Standards.

Mitsubishi Ethernet 2nd Generation (Q, L, iQR and iQF) Driver - MITS1

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

Property	Value
Error handling	
Retries	3
Disconnect after every transaction	False
Disconnect after every failure	False
Latency	
Inter station latency	0
Transaction latency	0
TXtoRX latency	0
Ethernet	
IP address	10.10.0.11
IP port	1280
Connect timeout	4000
Transaction timeout	500
Channel count	1
PLC details	
Max PDU	480
Other station enabled	False
MC network	0
MC station	0
CPU number	Local
CPU timeout	1000
Debug level bitmask	480
Stale threshold	0
Use custom ranges	True
PLC model	iQF CPU(FX5)
Test	
Ping size	32
Ping	-
Driver send watchdog	
Enable	False
Address	M 100
Rate (mSec)	5000
Type	LatchON
Pulse active time (mSec)	5000
Validation	
Use deprecated mechanism	False
Data ranges for selected PLC model (*WARNING: Denoted in DECIMAL below)	
X	1026
Y	1024
M	32768
L	32768
F	32768
B	32768
D	8255



Input relay - this is in DECIMAL notation here, but when scanning use HEX for all models except iQF_CPU(FX5) which must be in OCTAL notation.

Close

3. Device Configuration

3.1. Device Configuration

Enable secondary – Turn this option to have the driver maintain 2 channels of communication, the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

Enable shadow scanning from master to standby – NASA specific function – per Device Agent level shadow scanning.

If enabled, writes will be sent to the master to be executed on its corresponding device and reads from master will be replicated to the standby.

If disabled writes **Parallel scanning** is enabled.

Enable standby reads – If disabled, the standby will be considered cold, no reading will take place on the standby connection, and neither will any replication from the master occur.

Enable standby writes – If disabled, no writing will take place on the standby connection, and neither will any request be forwarded to the master.

3.1.1. Error Handling

Retries - (Default = 3) Number of attempts per transaction.

Disconnect after every transaction – (Default = False) -Tell the channel to close the port after every transaction. This can be useful on gateways if required.

Disconnect after every error – (Default = False) Tell the channel to close and reopen the port on every error during retries else it will only occur after the final retry.

3.1.2. Latency

Inter-PLC latency – The period that the system will wait before transmitting any read request or write request in multi-dropped system when a new device is contacted.

Transmission latency – The period that the system will wait before transmitting any read request or write request for the same plc device. Inter message delay.

Tx to RX latency – The period that the system will wait after transmitting a request and before attempting to read the response.

3.1.3. Ethernet

IP Address – IPv4 address. For example, 192.168.3.50

IP Port – (Default = 1280) For multiple channels a CSV format is supported. For example, say for Channels = 3, IP Port = 1280,1282,1289



PLEASE NOTE

Please ensure that the number of channels matches the number of ports listed in the CSV formatted. Please also ensure that the delimiter in the CSV is correct using a comma (`,`).

Connection Timeout – (ms) the time allowed for a connection to be established.

Transaction timeout – (ms) the time allowed for a completion of a transaction.

Channel count – Up to 8 simultaneous connections where the scan and write tasks will be load balanced on, interlaced (read, write, read ...)

3.1.4. PLC Details

Max PDU – this specifies the maximum number of words (16-bit) that the driver will allow to be transacted in a single transaction.

Other station enabled – (Default = False) Allows for a MELSEC network to access other CPUs remotely through this PLC.

MC Network – Network number of networks to be used.

MC Station – Remote PLC station number.

CPU Number – Enables communication to up to 4 CPU units on the same PLC. Also allows for selection of "Redundant" CPU pair and if selected will ensure that all communications is transferred to MASTER CPU in configured redundant system.

CPU timeout – The timeout setting to be used internally by the external card for transactions with the CPU. The actual value used is rounded up in increments of 250ms. Legal values are 1 through 63750ms.

Debug level bitmask – (Default = 480) This is a bitmask that enable extra debugging in the datascope output, do not leave it enabled during normal operation on high-speed drivers as it will slow down the system depending on speed requirements. In serial drivers this delay is negligible.



INFORMATION

Remember, all NASA drivers one can stop and start the device to reload these flags.

960 default is just the protocol messages 40hex + 80hex+ 100hex + 200hex = 3C0hex -> 960 decimal in the dialogue.

```
#define DEBUG_MASK_COMSRESOURCE_INFO          0x00000001
#define DEBUG_MASK_COMSRESOURCE_CRITICAL      0x00000002
#define DEBUG_MASK_PROTOCOL_ITEM_VALUES       0x00000004
```



```
#define DEBUG_MASK_PROTOCOL_ITEM_BAD_VALUES 0x00000008
#define DEBUG_MASK_NASA_POOLS 0x00000010
#define DEBUG_MASK_TIMING 0x00000020
#define DEBUG_MASK_PROTOCOL_RESPONSES 0x00000040
#define DEBUG_MASK_PROTOCOL_REQUESTS 0x00000080
#define DEBUG_MASK_PROTOCOL_READ_DESC 0x00000100
#define DEBUG_MASK_PROTOCOL_WRITE_DESC 0x00000200
```

Stale threshold – (Default = 0) Every nth occurrence of an unchanged item value will be sent to the master. If the value is different, the value will be sent regardless of this count. This count reduces unnecessary event within the Agent Server.

Use custom ranges – Default – FALSE, if set to TRUE the device will override any or all type ranges defined below.



WARNING

Some I/O types are denominated in HEX, DECIMAL or OCTAL , the ranges below are denominated in DECIMAL but may be another Type when referenced during the scanning phase. E.g. Selecting the iQF CPU(FX5) and adjusting the X_TYPE range to 1026 (decimal!) will allow addresses to scan up to including X 2002 which is 1026 decimal. (Calc.exe in programmer mode is your friend).

Overriding any ranges needs to match what is actually available in the target plc.

PLC Model – Select the relevant PLC CPU model; Q Series CPU, QnA Series CPU, L Series CPU, R Series CPU, FX5(iQF) Series By selecting a CPU, the device address range is set. Custom allows you to modify these ranges. Selecting a new model when Use custom ranges is FALSE, will cause the TYPE ranges below to be updated to their defaults, else if Use custom ranges is TRUE the TYPE ranges will be left as is so as to preserve the required new size counts.

3.1.5. Test

Ping size – (Default = 32 bytes) Adjustable but not more than 65000 bytes. Indicates the number of bytes to send with the ICMP ping packet. This is only an indicator of IP destination acknowledgment during the life of the setup dialog.

Ping - (and result) Click the test button on the far left.

3.1.6. Driver Send Watchdog

This is an internal driver based write to help the PLC program to check if the communications between master and PLC is healthy. It can be used to pause operations or cycle the network, etc.

Enable – (Default = OFF) Enable to tell the device to generate an internal write based on **Type**, to a specified bit **Address**.

Address – (Default = M 100) Any writeable bit Address.

Rate (mSec) – (Default = 2000ms) The period the driver will initiate a new watchdog send.

Type – (Pulse on/off, Latch on/off) – Pulse will generate a configured on or off per the **Pulse active** time duration.

Pulse active time (mSec) – (Default = 5ms) if pulse type is used, this will be the pulse width. It is recommended to keep it much smaller the **Rate**. If Latch is used, this will be ignored.

3.1.7. Validation

Use deprecated mechanism – (Default = False) If set to true, it will revert the scanning validation algorithm to the older method. For Backward compatibility only.



WARNING

This deprecated mechanism option is here for backward compatibility use only , do not use it on a new plc installation.
Changing this on an existing plc might cause changes in address interpretation, set to true only if one suspects problems with scanning, contact Adroit Drivers for assistance if required.
As with any change of this nature make sure relevant backups are kept before and after a change.
(WGP file)

3.1.8. Data Ranges for Selected PLC Model (*WARNING: Denoted in DECIMAL below)

X,Y ...ZR - This section specifies the maximum address that can be validated for each address type.



WARNING

WARNING: These ranges are denominated in DECIMAL for display purposes only.
See the description box below to understand the actual notation the driver will be expecting and use, the driver will reject invalid addressing, e.g. 8 and 9 not used in octal, A-F not used in decimal nor octal.

Selecting the iQF CPU(FX5) and adjusting the X_TYPE range to 1026 (decimal!) will allow addresses to scan up to including X 2002 octal (1026 decimal). (Calc.exe in programmer mode is your friend)

4. Supported Addresses

4.1. Mitsubishi Q series (System Q)

Device	Prefix	Max Range	Notation	PLC Data Type	Boolean	Integer	Real	String
Inputs	X	0x1FFF	HEX	Bit	X 0	-	-	-
				Word16	-	X 0 W	-	-
Output relay	Y	0x1FFF	HEX	Bit	Y 0	-	-	-
				Word16	-	Y 0 W	-	-
Latch relay	L	8191	DEC	Bit	L 0	-	-	-
				Word16	-	L 0 W	-	-
Internal relay	M	8191	DEC	Bit	M 0	-	-	-
				Word16	-	M 0 W	-	-
Link relay	B	0x1FFF	HEX	Bit	B 0	-	-	-
Special link relay	SB	0x7FF	HEX	Bit	SB 0	-	-	-
Edge relay	V	2047	DEC	Bit	V 0	-	-	-
Data register	D	12287	DEC	Bit	D 0.0 D 0.F	- -	- -	- -
				Int16 (def.)	-	D 0 I	D 0 I	D 0 I
				Word16	-	D 0 W	D 0 W	D 0 W
				DWord32	-	D 0 D	D 0 D	D 0 D
				Long32	-	D 0 L	D 0 L	D 0 L
				Float32	-	-	D 0 F	D 0 F
				String/Text	-	-	-	D 0 Sn
Link register	W	0x1FFF	HEX	Bit	W 0.0 W 0.F	- -	- -	- -
				Int16	-	W 0 I	W 0 I	W 0 I
				Word16	-	W 0 W	W 0 W	W 0 W
				DWord32	-	W 0 D	W 0 D	W 0 D
				Long32	-	W 0 L	W 0 L	W 0 L
				Float32	-	-	W 0 F	W 0 F
				String/Text	-	-	-	-
Timer (current value)	T or TN*	2047	DEC	Int16	-	TN 0 I	TN 0 I	-
Retentive timer (cur. val)	ST	2047	DEC	Int16	-	ST 0 I	ST 0 I	-
Counter (current value)	C or CN*	1023	DEC	Int16	-	CN 0 I	CN 0 I	-
Special link reg.	SW	0x7FF	HEX	Int16	-	SW 0 I	SW 0 I	-
File register	R	32767	DEC	Bit	R 0.0 R 0.F	- -	- -	- -
				Int16	-	R 0 I	R 0 I	R 0 I
				Word16	-	R 0 W	R 0 W	R 0 W
				DWord32	-	R 0 D	R 0 D	R 0 D
				Long32	-	R 0 L	R 0 L	R 0 L
				Float32	-	R 0 F	R 0 F	R 0 F
				String/Text	-	-	-	R 0 Sn
Extended File register	ZR	4184063	DEC	Bit	ZR 0.0 ZR 0.F	- -	- -	- -
				Int16	-	ZR 0 I	ZR 0 I	ZR 0 I
				Word16	-	ZR 0 W	ZR 0 W	ZR 0 W
				DWord32	-	ZR 0 D	ZR 0 D	ZR 0 D
				Long32	-	ZR 0 L	ZR 0 L	ZR 0 L
				Float32	-	ZR 0 F	ZR 0 F	ZR 0 F
				String/Text	-	-	-	R 0 Sn

4.2. Mitsubishi QnA series PLC

Device	Prefix	Max Range	Notation
Inputs	X	0x1FFF	HEX
Output relay	Y	0x1FFF	HEX
Internal relay	M	8191	DEC
Latch relay	L	8191	DEC
Annunciator	F	2047	DEC
Link relay	B	0x1FFF	HEX
Special link relay	SB	0x7FF	HEX
Edge relay	V	2047	DEC
Data register	D	8191	DEC
Link register	W	0x1FFF	HEX
Timer (current value)	T or TN*	2047	DEC
Retentive timer (current value)	ST	2047	DEC
Counter (current value)	C or CN*	1023	DEC
Special link reg.	SW	0x7FF	HEX
File register	R	32767	DEC
Extended File register	ZR	524287	DEC

4.3. Mitsubishi L series PLC data

Device	Prefix	Max Range	Notation
Inputs	X	0x7FFF	HEX
Output relay	Y	0x7FFF	HEX
Internal relay	M	8191	DEC
Latch relay	L	8191	DEC
Annunciator	F	2047	DEC
Link relay	B	0xFFFF	HEX
Special link relay	SB	0x7FF	HEX
Edge relay	V	2047	DEC
Data register	D	6143	DEC
Link register	W	0xFFFF	HEX
Timer (current value)	T or TN*	2047	DEC
Retentive timer (current value)	ST	2047	DEC
Counter (current value)	C or CN*	1023	DEC
Special link reg.	SW	0x7FF	HEX
File register	R	32767	DEC
Extended File register	ZR	524287	DEC

4.4. Mitsubishi R series PLC data

Device	Prefix	Max Range	Notation
Inputs	X	0x2FFF	HEX
Output relay	Y	0x2FFF	HEX
Internal relay	M	12287	DEC
Latch relay	L	8191	DEC
Annunciator	F	2047	DEC
Link relay	B	0x1FFF	HEX
Special link relay	SB	0x7FF	HEX
Edge relay	V	2047	DEC
Data register	D	18431	DEC
Link register	W	0x1FFF	HEX
Timer (current value)	T or TN*	1023	DEC
Retentive timer (current value)	ST	0	DEC
Counter (current value)	C or CN*	511	DEC
Special link reg.	SW	0x7FF	HEX
File register	R	32767	DEC
Extended File register	ZR	4184063	DEC

4.5. Mitsubishi FX5 series PLC data

Device	Prefix	Max Range	Notation
Inputs	X	337	OCT
Output relay	Y	337	OCT
Internal relay	M	8255	DEC
Latch relay	L	0	DEC
Annunciator	F	0	DEC
Link relay	B	0	HEX
Special link relay	SB	0	HEX
Edge relay	V	0	DEC
Data register	D	8255	DEC
Link register	W	0	OCT
Timer (current value)	T or TN*	255	DEC
Retentive timer (current value)	ST	0	DEC
Counter (current value)	C or CN*	255	DEC
Special link reg.	SW	0	OCT
File register	R	0	DEC
Extended File register	ZR	0	DEC

If the PLC's memory range is smaller than the range supported by the driver, it is possible to configure the driver to scan non-existent I/O addresses. Since this can cause unpredictable results, please ensure that all scanned I/O are valid for your controller's current configuration.

* Timer/Counter current values are supported, but the contact and coil values are not scanned. T and C prefixes are promoted to TN and CN respectively.

Adroit string slots and ASCII type is only supported to Data/File registers and can be scanned as "D/R 0 Sn" where n is the number of ASCII characters in the string required, which can go up to a maximum of 79 characters.

4.6. Omitted

The following I/O types or devices are not supported:

Device	Mnemonic
Function in/outputs	
Timer and counter coils and contacts	TS, TC, SS,SC, CS,CC
Step relay	S
Direct in/out	DX,DY

4.7. Non-bit Device Writes

Bit-writes to items that are not supported in the protocol as discrete (bit devices) will be supported using *read-mask-write*, at a cost of transaction speed.

Bit Devices:

Device	Prefix	Base
Input relay	X	HEX
Output relay	Y	HEX
Internal relay	M	Decimal
Latch relay	L	Decimal
Annunciator	F	Decimal
Link relay	B	HEX
Special link relay	SB	HEX
Edge Relay	V	Decimal

1. Bit devices are always accessed in groups of 16, except when writing to a bit.
2. When addressing Integer and Real slots, it is possible to address 32-bit devices in a grouping. A suffix to the scan-address must also be supplied to indicate the size and presentation.

Supported suffixes for Integer/Real Adroit slots:

Unsigned 16-bit	W
Signed 16-bit (default)	I
Unsigned 32-bit	D
Signed 32-bit	L
Single precision 32-bit float	F

Word Devices:

Device	Suffix	Base	Boolean	Integer	Real
Data register	D	Decimal	D0.0	D0	D0
Link register	W	HEX	W0.0	W0	W0
Timer (current value)	T or TN*	Decimal	none	TN0	TN0
Retentive timer (current value)	SN	Decimal	none	SN0	SN0
Counter (current value)	C or CN*	Decimal	none	CN0	CN0
Special link reg.	SW	HEX	SW0.0	SW0	SW0
File register	R	Decimal	R0.0	R0	R0
Extended File register	ZR	Decimal	ZR0.0	ZR0	ZR0

1. When accessing a word device in a Boolean or Digital tag, a bit number from 0 through 15 must be supplied.
2. Only when accessing a word device in a Real or Analog tag, does the floating-point presentation become available, (suffix F).
3. Timers and counter types do not support float "F" presentation.
4. Extended file registers are not supported.

5. Protocol Definition and Considerations

5.1. PLC software and Heartbeat code

5.1.1. PLC Side Communication setup

Two possible configurations are presented, one without a watchdog timer, and one with a watchdog. The driver thus supports two modes: one with heartbeat, and one without. PLC logic is required for the watchdog. If a watchdog/heartbeat is not used, the AJ must simply be configured using GX developer. Sample code using IEC to configure the AJ is not supplied. Any sample programs are supplied as-is and are not under any warranty.

5.2. Using GX Works2 (Used for All except **R Series** CPU)

5.2.1. Settings for CPU without an on-board Ethernet Connection (Q02)

Operational Settings

In the Ethernet Network Parameters of the PLC Program specify the Ethernet module in the required slot and then select "Operation Setting". Modify settings based on the image below with the Ethernet modules IP Address in it.

Ethernet Operation Setting
✕

Communication Data Code

☒ Binary Code

☐ ASCII Code

Initial Timing

☐ Do not wait for OPEN (Communications impossible at STOP time)

☒ Always wait for OPEN (Communication possible at STOP time)

IP Address Setting

Input Format: DEC

IP Address: 192 0 1 254

Send Frame Setting

☒ Ethernet(V2.0)

☐ IEEE802.3

☒ Enable Online Change

TCP Existence Confirmation Setting

☒ Use the KeepAlive

☐ Use the Ping

End
Cancel

Open Settings

Configure the connections as below (note how the port address 1280 is in decimal format). Copy the connection as many times as needed for additional ports.

IP Address/Port No. Input Format DEC									
	Protocol	Open System	Fixed Buffer	Fixed Buffer Communication	Pairing Open	Existence Confirmation	Host Station Port No.	Destination IP Address	Destination Port No.
1	TCP	Unpassive	Send	Procedure Exist	Disable	Confirm	1280		

Save, and download the program (you may need to cycle power on the PLC).

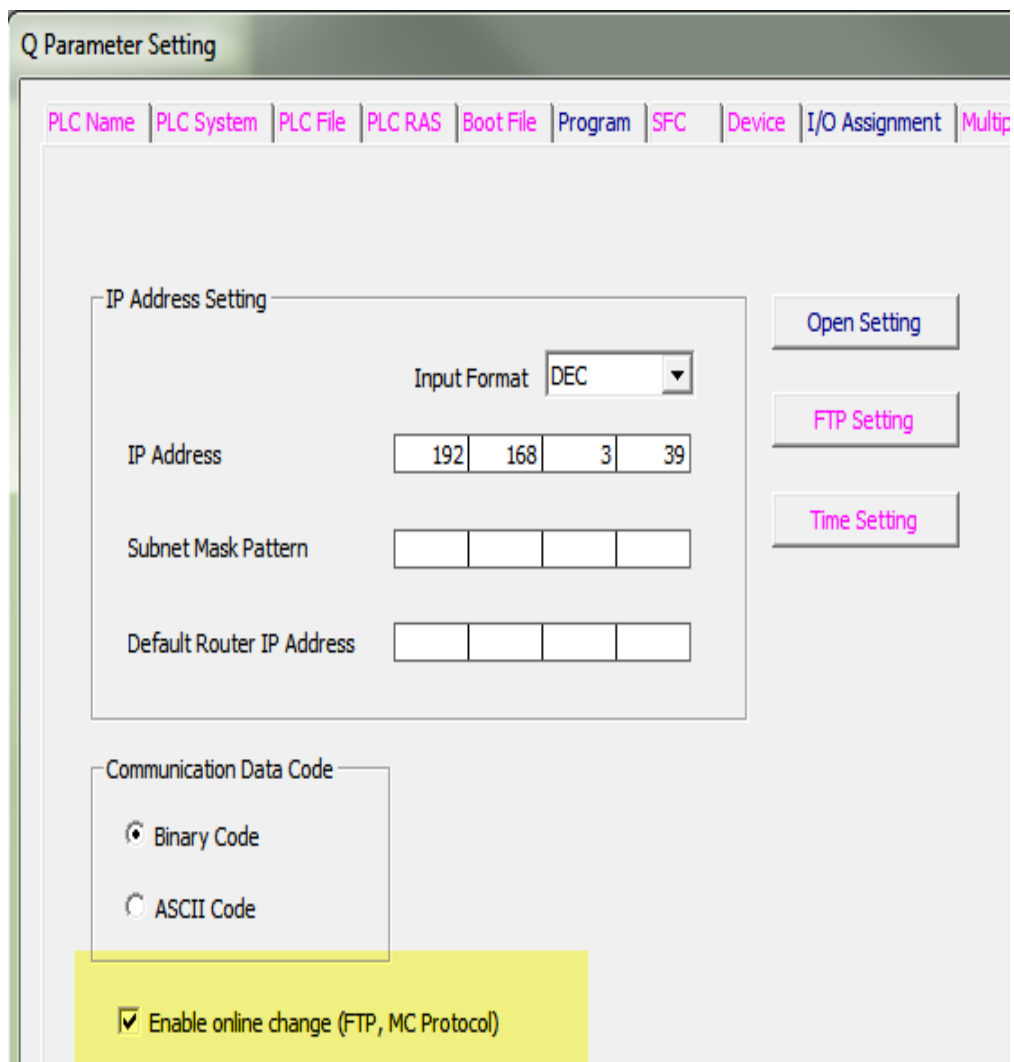
Initial Settings

The default initial settings work very well, however, to more easily recover broken network connections, the confirmation existence option above can be changed to "confirm", and the set of initial settings used in the watchdog section may also be used. Using "Confirm" connections uses a TCP/IP keep-alive, which will cause the PLC to detect and re-open a port in some of the possible network failure conditions.

5.2.2. Settings for CPU with an on-board Ethernet Connection (Q03UDE)

Operational settings

In the PLC Parameters settings of the PLC Program is a TAB called "Build-in Ethernet Port Setting". Select this TAB and specify the Ethernet module settings based on the image below with the Ethernet modules IP Address in it.



Q Parameter Setting

PLC Name | PLC System | PLC File | PLC RAS | Boot File | Program | SFC | Device | I/O Assignment | Multiple

IP Address Setting

Input Format: DEC

IP Address: 192 | 168 | 3 | 39

Subnet Mask Pattern: | | | |

Default Router IP Address: | | | |

Communication Data Code

☒ Binary Code

☐ ASCII Code

☒ Enable online change (FTP, MC Protocol)

Open Setting

FTP Setting

Time Setting

Open Settings

Configure the connections as below (note how the port address 0500 is in hexadecimal format for 1280) Copy the connection as many times as needed for additional ports.

IP Address/Port No. Input Format DEC

	Protocol	Open System	TCP Connection	Host Station Port No.	Destination IP Address	Destination Port No.
1	UDP	MELSOFT Connection				
2	TCP	MELSOFT Connection				
3	TCP	MELSOFT Connection				
4	TCP	MELSOFT Connection				
5	TCP	MELSOFT Connection				
6	TCP	MELSOFT Connection				
7	TCP	MELSOFT Connection				
8	TCP	MELSOFT Connection				
9	TCP	MELSOFT Connection				
10	TCP	MELSOFT Connection				
11	TCP	MELSOFT Connection				
12	TCP	MELSOFT Connection				
13	TCP	MELSOFT Connection				
14	TCP	MELSOFT Connection				
15	TCP	MELSOFT Connection				
16	TCP	MC Protocol		1280		

(*) IP Address and Port No. will be displayed by the selected format.
 Please enter the value according to the selected number.

End
Cancel

Save, and download the program (you may need to cycle power on the PLC).

Initial Settings

The default initial settings work very well, however, to recover broken network connections, the confirmation existence option above can be changed more easily to "confirm" and the set of initial settings used in the watchdog section may also be used. Using "Confirm" connections uses a TCP/IP keep-alive, which will cause the PLC to detect and re-open a port in some of the possible network failure conditions.

5.3. PLC Setup with PLC-SCADA Watchdog

The watchdog mode allows the driver to recover from the following scenario: unplug network in the middle of a transaction; wait 5 seconds; reconnect. In this scenario, the driver thinks the PLC has disconnected, and re-opens the connection, but the PLC does not know the PC is having problems, and thus denies the new connection, since it only allows one connection at a time.

5.3.1. Initial Settings

Timer setting

Module will operate with default values if setting is left blank

	Setting value	Default value	In units
TCP ULP timer	20	60	X500ms
TCP zero window timer	4	20	X500ms
TCP resend timer	4	20	X500ms
TCP end timer	12	40	X500ms
IP assembly timer	3	10	X500ms
Response monitoring timer	30	60	X500ms
Destination existence conformation starting interval	10	1200	X500ms
Destination existence conformation interval timer	10	20	X500ms
Destination existence conformation resend timer	2	3	Times

DNS setting

Input format: DEC.

IP address of DNS server 1				
IP address of DNS server 2				
IP address of DNS server 3				
IP address of DNS server 4				

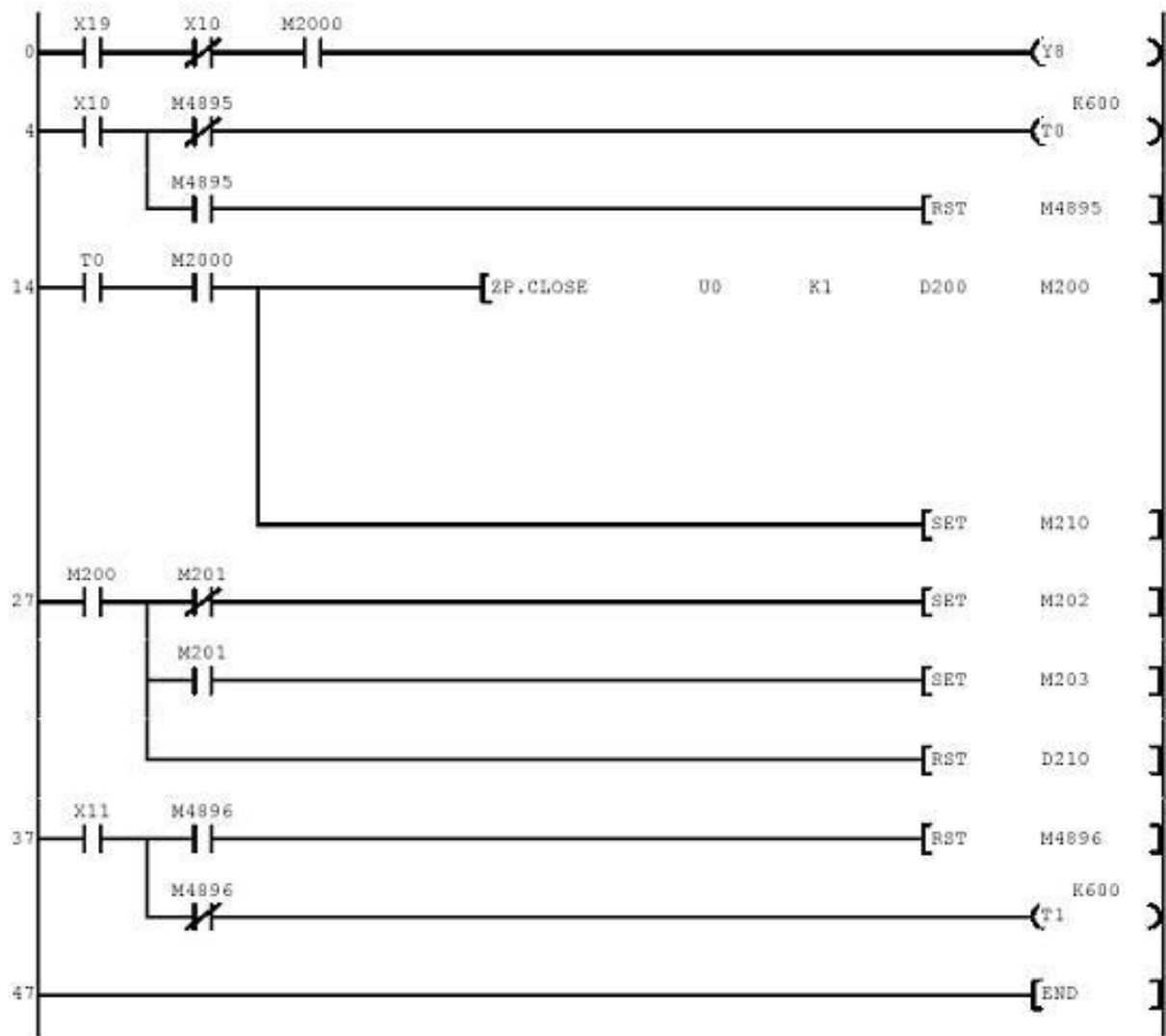
The timeouts have been reduced in order to detect the connection break sooner.
Save, and download the program (you may need to cycle power on the PLC).

5.3.2. PLC Watchdog Program

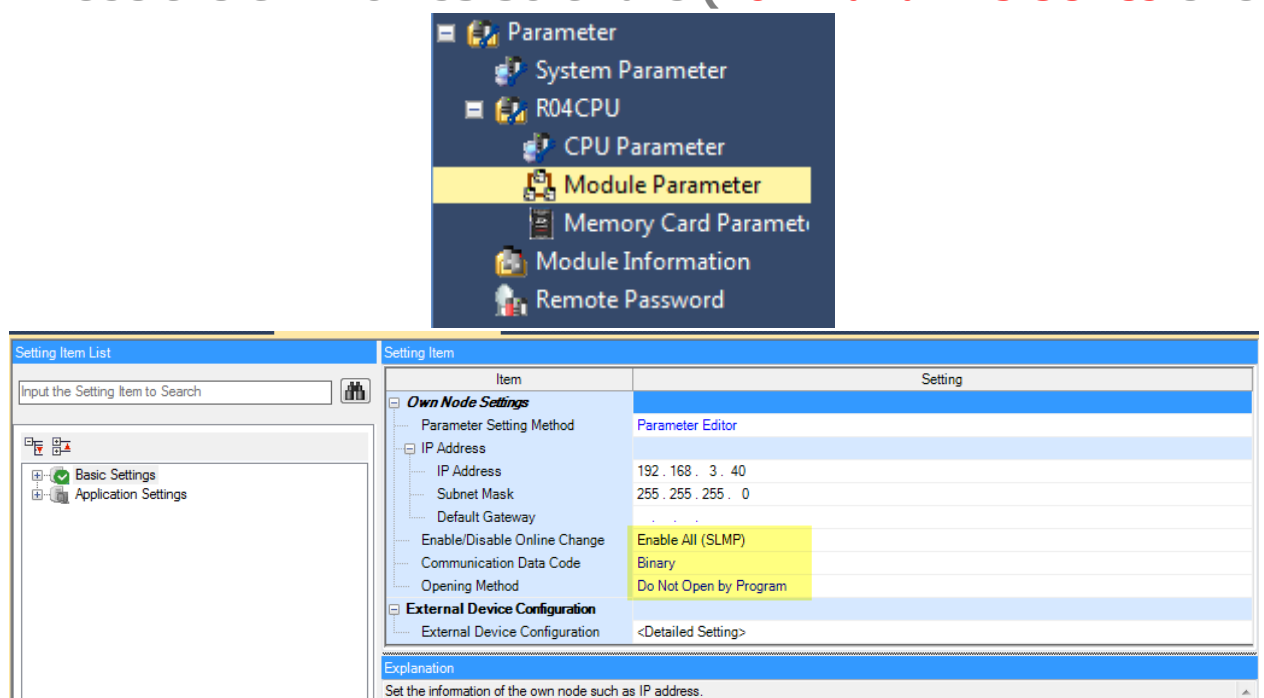
Additionally, the watchdog portion of the program used for the A-Series controller may be used to ensure that connections which are terminated unexpectedly due to hardware faults will still be recovered by the AJ on the System-Q. This operates on the basis of the program running a timer for the watchdog bit while off, when the watchdog bit turns on, the timer is reset. If the timer exceeds a count equal to twice the watchdog interval specified in the Adroit driver, the PLC will shut the connection and allow it to be re-opened by the Adroit driver again.

Since the driver keeps turning the bit on periodically, the timer should never expire in normal operation. When it does, the PLC program knows the PC was unable to set the bit, because it cannot or it is not communicating. One timer and watchdog bit is allocated per PLC TCP/IP port. Please see reference for the AJ for I/O numbers for the consecutive ports.

The PLC Program required if a watchdog is used is shown below. This program shows two ports being maintained.

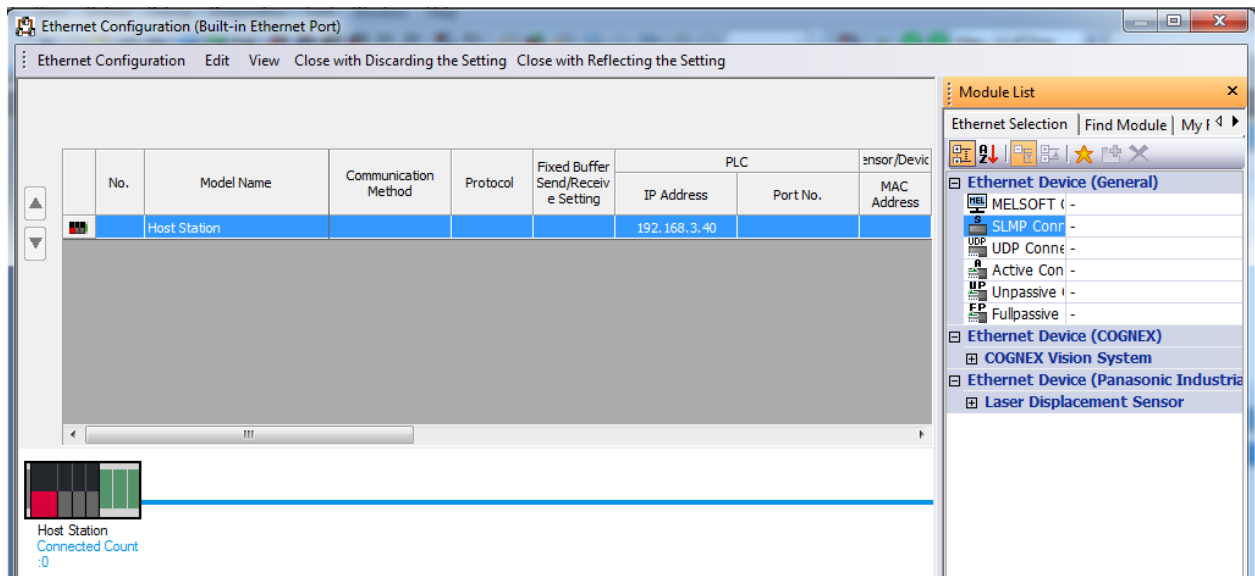


5.4. Use the GX Works3 Software (For R and FX5 Series CPU's)

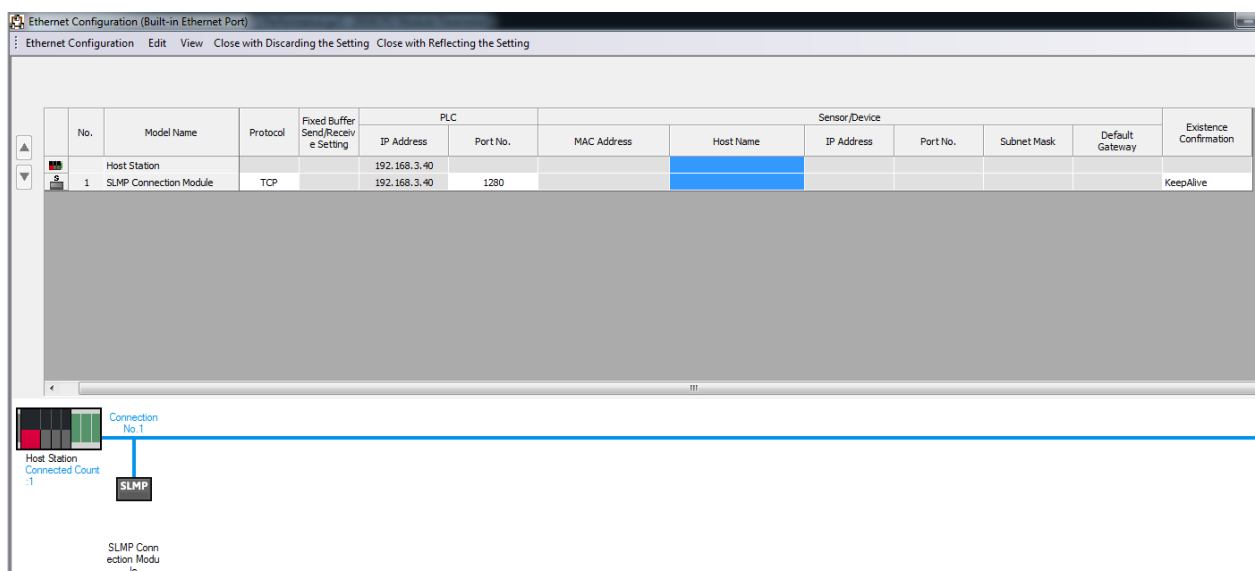


Enter IP address of the module and also do ALL settings marked as yellow in the above Screenshot.

Then select the External Device configuration, Detailed Setting:



Drag and Drop a SLMP connection into the Main window.



Specify the TCP protocol, Port No. 1280 decimal and "keepAlive" existence check. Download those parameters to the PLC and reset the PLC. Done.

5.5. The MC Protocol 3E Frame [Binary]

5.5.1. Request-Frame Format

All multi-byte fields are ordered L-H (LSB first) in binary, and MSB first in ASCII frames.

Header	Q Header						Text
	Network no.	PC no.	Dest module I/O no.	Dest module station no.	Request length	CPU Timer	
50 00	00	FF	FF 03	00	20 00	10 00	

Request length is # bytes from this field onwards in the frame, for ASCII frames, this is the total number of characters in subsequent fields. For binary frames, this is the number of bytes.

CPU timer is in 250ms increments 16(dec) = 4000 ms.

5.5.2. Response-Frame Format

Header	Q Header						Text
	Network no.	PC no.	Dest module I/O no.	Dest module station no.	Request length	Response code	See below
50 00	00	FF	FF 03	00	20 00	00 00	

5.5.3. Text Portion of the 3E/3C Frame

Batch Read from word device (read "D2")

Command	Sub-command	Head device number	Device code	# of device points
04 01	00 00	02 00 00	A8	01 00

Batch Read from bit device (read bit "X2")

Command	Sub-command	Head device number	Device code	# of device points
04 01	00 01	02 00 00	9C	01 00

Batch Write to word device

Command	Sub-command	Head device number	Device code	# of device points	Data...
14 01	00 00	00 00 00	A8	01 00	XX XX

Batch Write to bit device (write "Y0")

Command	Sub-command	Head device number	Device code	# of device points	Data...
14 01	00 01	00 00 00	9D	01 00	XX XX

For an ASCII frame, the device-Head and Device-Code (4 bytes) are reversed

5.5.4. Device Codes

Device	ASCII Frame	BINARY Frame
Special relay ♀	SM	91
Special register ♀	SD	A9
In	X*	9C
Out	Y*	9D
Internal relay	M*	90
Latch relay	L*	92
Annunciator	F*	93
Edge Relay	V*	94
Link relay	B*	A0
Data register	D*	A8
Link register	W*	B4
Timer – contact ♀	TS	C1
Timer – coil ♀	TC	C0
Timer – current ♀	TN	C2
Retentive timer – contact	SS	C7
Retentive timer – coil	SC	C6
Retentive timer – current	SN	C8
Counter – contact ♀	CS	C4
Counter – coil ♀	CC	C3
Counter – current ♀	CN	C5
Special link relay	SB	A1
Special link register	SW	B5
Step relay ♀	S*	98
Direct input ♀	DX	A2
Direct output ♀	DY	A3
Index register ♀	Z*	CC
File Register	R*	AF
	ZR	B0

♀ Types with ♀ are not supported by the driver

* may use space or * in ASCII protocol frame

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting Steps

1. The RUN and INIT lamps on the AJ must be lit.
2. The COM.ERR lamp indicates a protocol or hardware fault has occurred.
3. Ping the relevant PLC Ethernet card E.g.
ping 192.168.1.11
The ping should be echoed back.
4. During a ping, the RD and SD lamps on the external module will flicker together.
5. Check that the port is not already open, if another device is already connected to the PLC, the OPEN lamp will be lit. It is possible that multiple equipments are connected and have an open port. To see the status of open ports, the "Ethernet diagnostics" menu in GX Developer.

6.4. Error Message Reference:

Only commonly encountered PLC error codes are listed, for a full reference, see Melsec Q Programmable Controller User manual (basic) QJ71E71. The PLC error codes can be seen in the Adroit event log, alarm viewer and driver Datascope screens, where a reason description also appears.

HEX Code	Remedial Action
0	No error
4031	The scanned device or I/O does not exist. Verify PLC device range configuration or existence of File register.
C050	Configure the Q71 card for binary and restart the module.
C051 - C055	The number of points to read/write is incorrect, contact Adroit technical support.
C056	The scanned device or I/O does not exist. The starting address + length of the request exceeds the number of points in the PLC. Verify PLC device range configuration.
C057	Requested data length miss-match, contact Adroit technical support.
C058	Requested data length miss-match, contact Adroit technical support.
C059	The command issued is invalid, contact Adroit technical support.
C05B	The requested device does not exist. Verify PLC device configuration.
03E8 - 4FFF	Are generated by the CPU, see CPU error code manual for more detail.

In the event of remedial action failing to rectify a problem, the context of the error message is needed by our team. Please see the Adroit help file on how to capture driver data screens by searching for "to create a protocol dump" (no quotes) in the search tab.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.2	19-Jan-2023	K. Robinson	Adjusted grammar, image alignments	D. Jordaan
2.0	19-Aug-2025	D Jordaan	Added new information upto Rev 55.	
3.0				
4.0				
5.0				
6.0				

7.2. DLL Revision

Rev.	Date	Changes
37	05/02/2020	Document updated for Rev 10.0.4.1
46	18/01/2021	Document updated for Rev 10.0.5.x
49	18/05/2021	Document updated for Rev 10.0.5.1 Address validation strengthened as there were some addressing scenarios that did not scan correctly, the old, deprecated method was kept but deprecated and allowed as an opt-in to the configuration for backward compatibility only. New : Strings now correctly prompt for the correct next address when bulk scanning. Addresses now correctly prompts multi-register items (L/D/F/Sn) that could overlap the end register in a device area. Added Detailed Address information for the scanning details button.
50	30/11/2022	Reformat and update
53	30/11/2022	Shadow mode partner (with offline redundant plc) in standby did not mark affected tags bad when becoming master. For Shadow and Parallel, when switching cluster : unchanging tags were not updated , first update values now also work as expected.
54	26/04/2022	Short malformed scan addresses can be rejected, e.g. "R0" was not correctly formed into "R 0 I" etc... Fixed
55	19/08/2025	Added "Use custom ranges" , updated examples and warning around using custom ranges

